Conditional zero order at an actual projective branch

Formal proof packet

27 October 2026

This packet records a conditional local-geometry proof, not a target experiment or cost comparison. The editable diagram, gate CSV, Lean source, and compiler log are beside this PDF.

Checked statement

Let a be a rational simple root of the quintic and R the structure-sheaf stalk of the actual weighted-Proj curve at that branch. The previously checked local geometry makes R a DVR, with $\mathrm{ord}(y) = 1$ and $y^2 = (x - a)q$ for a unit q. The new theorem proves that if $N = (x - a)t$ and B is a unit in R, then

$$\mathrm{ord}(N - By) = 1.$$

The proof transports the generic branch zero-order lemma into this actual projective stalk. The underlying DVR criterion is discussed in the Stacks Project, Lemma 10.119.7.

A second checked theorem composes the separate norm-factor transport with this result. If a polynomial norm identity holds at a rational support root and B is a unit in this stalk, then the evaluated polynomial numerator satisfies $\mathrm{ord}(n(x) - By) = 1$. A checked corollary replaces the unit premise with the source-style scalar guard $b \neq 0$: its image B is a unit. The norm polynomial and its link to the compiled runtime numerator remain explicit inputs.

The source-style specialization uses the checked common 2+2 coefficient formulas. Under valid Mumford input congruences, a normalized quintic head, nonzero determinant, a rational simple branch, and a root of the norm support product, it proves the same local order for the proof-side polynomial numerator. Equality of the supplied quintic with the actual curve equation and compiled runtime semantics are still separate obligations.

The actual-quintic specialization proves the weighted-Proj quintic evaluates to the affine branch polynomial and has the required normalized head. A further checked corollary derives the simple-root condition from separability of that quintic. Valid input congruences, nonzero determinant, rational branch root, and norm-support root remain explicit.

Checked actual stalk: R is a DVR and $\mathrm{ord}(y) = 1$

Explicit input: $N = (x - a)t$ (or the checked norm route), $b \neq 0$

$\Downarrow$ checked local valuation algebra

$\mathrm{ord}(N - By) = 1$

Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero for all five sources. All thirteen theorem axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256 hashes are in the companion CSV and Markdown report; the updated sources are:

Quintic head: `668bd42e2f25d22dfa128cf910572f76190ec630a56665e8808b42bcf4d643f3.`

Actual specialization: `c7373682f8ce09e0c2243dfa44c92b9e6467f0b9188fa420aa1c23fbc4e6ca7d.`

The compiled Rust law has not been shown to equal the mapped polynomial on every branch. Nonbranch multiplicities, the global principal divisor, the Pic^0 class equation, and universal runtime semantics remain open. No frozen 1,260-target row or matched-rho ratio changes.